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Prn no : 202501040051

Division : CS-1

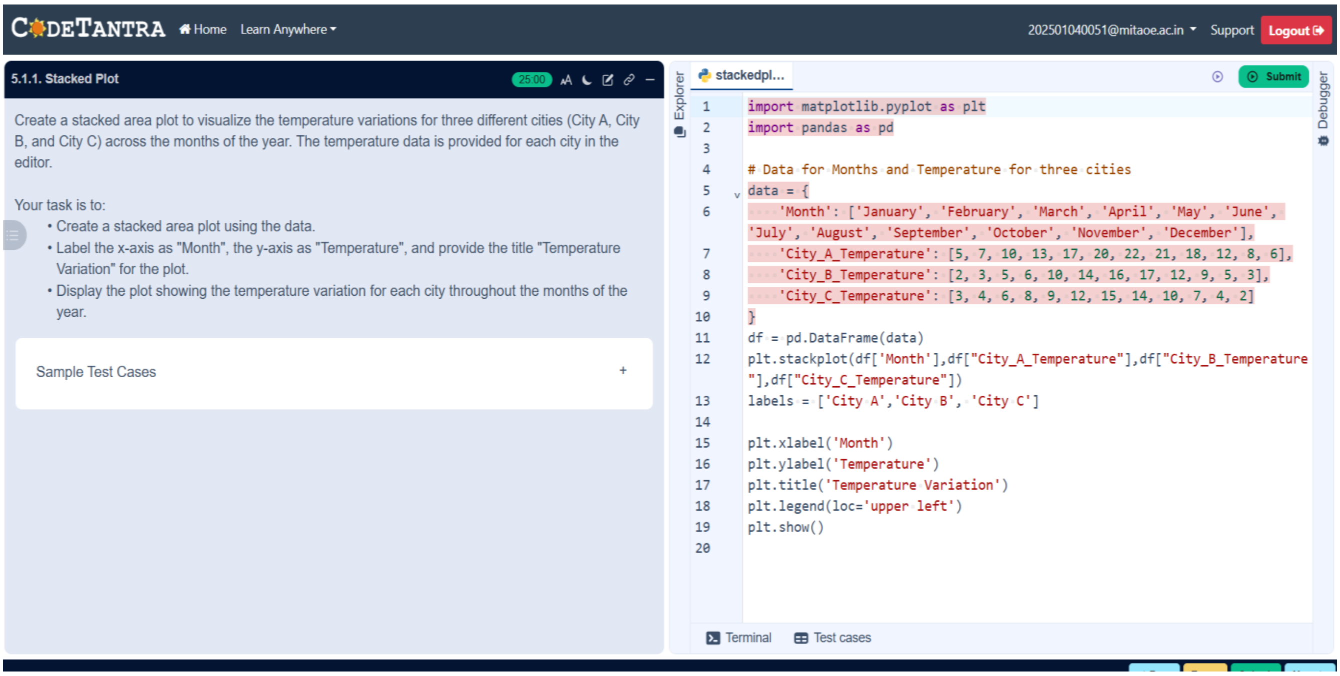
Batch : C13

Course : EDS

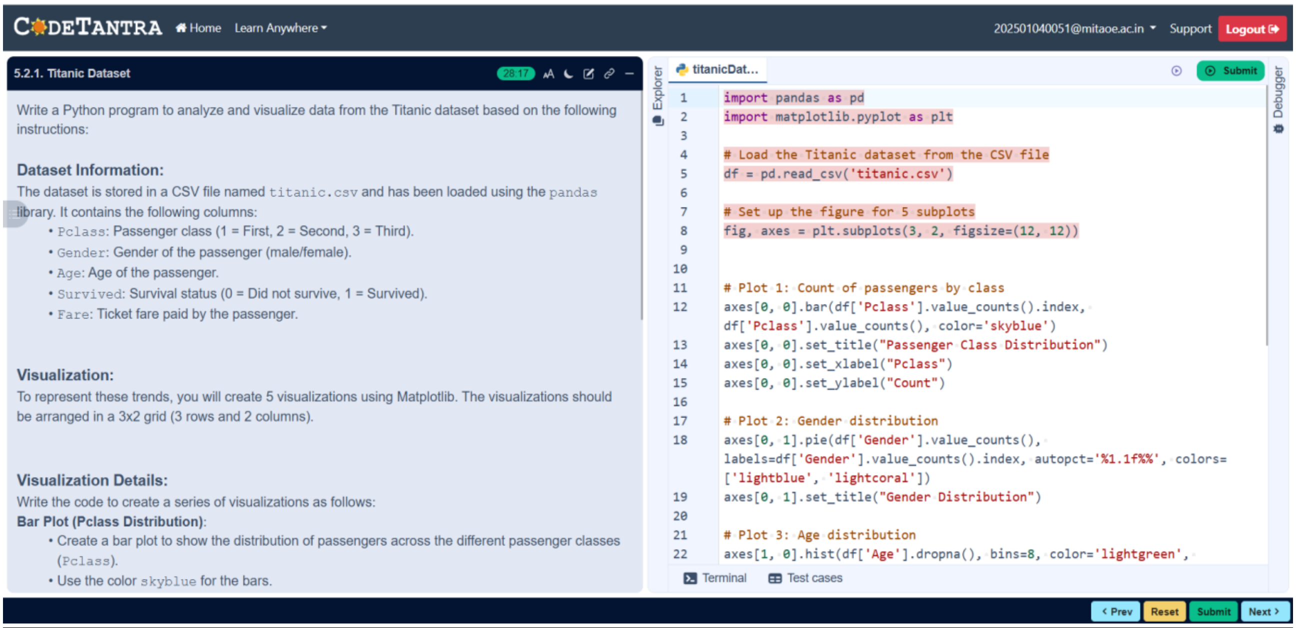
Essential of Data Science Laboratory

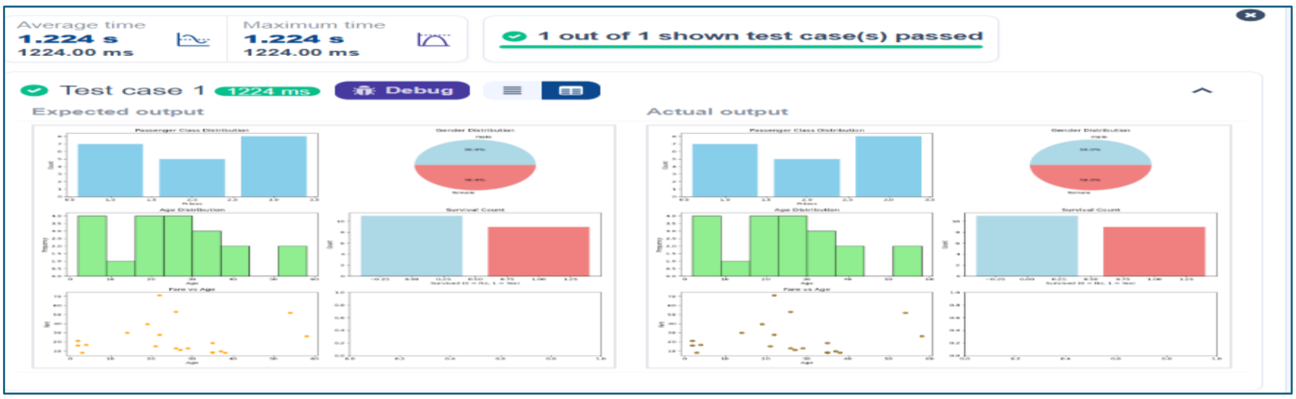
Practical 5:

5.1.1 Stacked Plot:

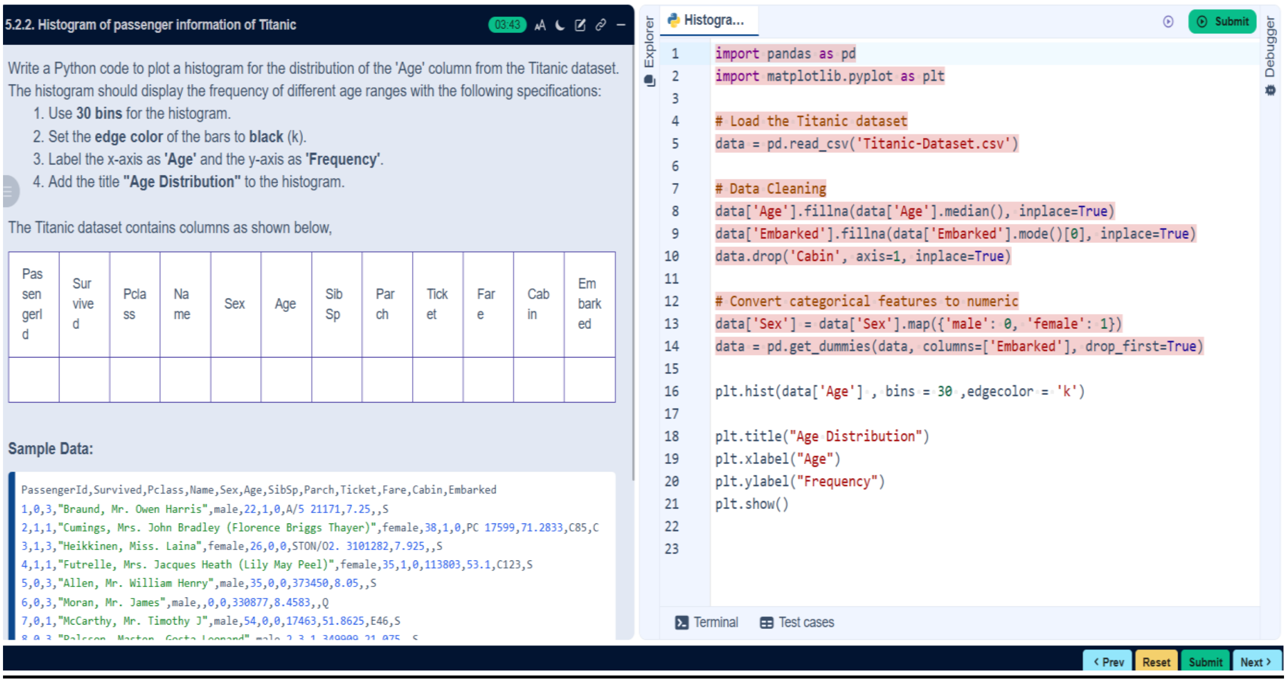


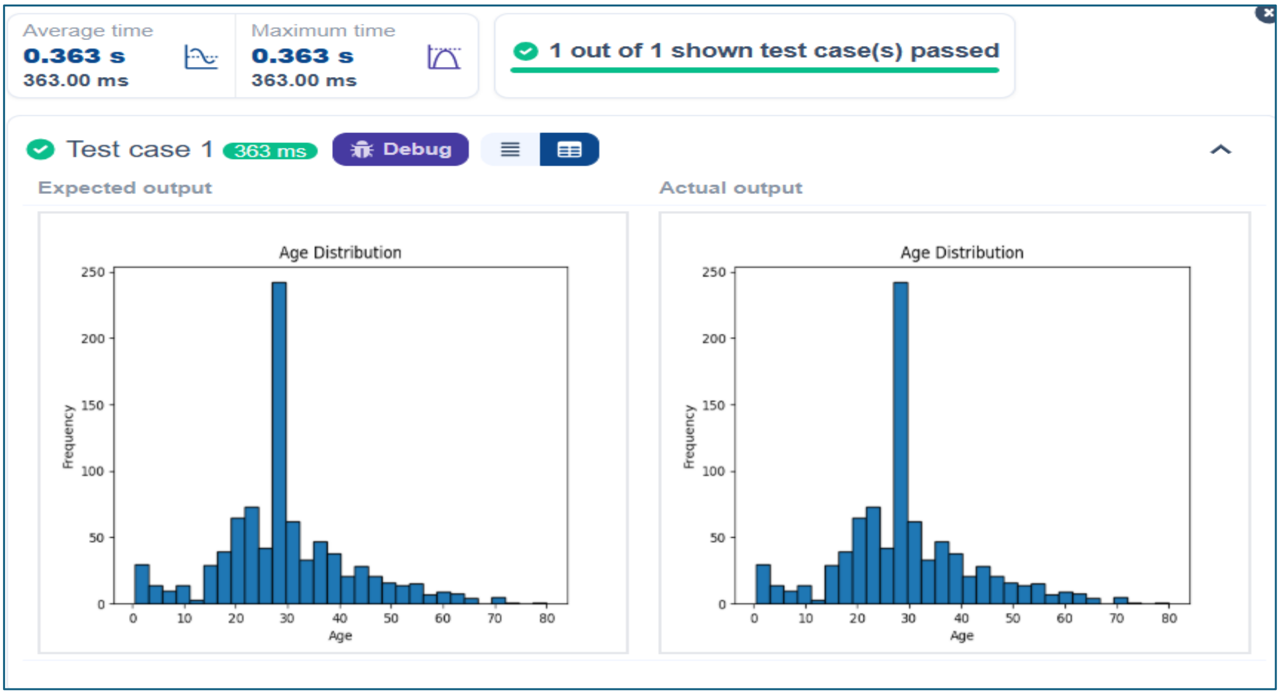
5.2.1 Titanic Dataset:



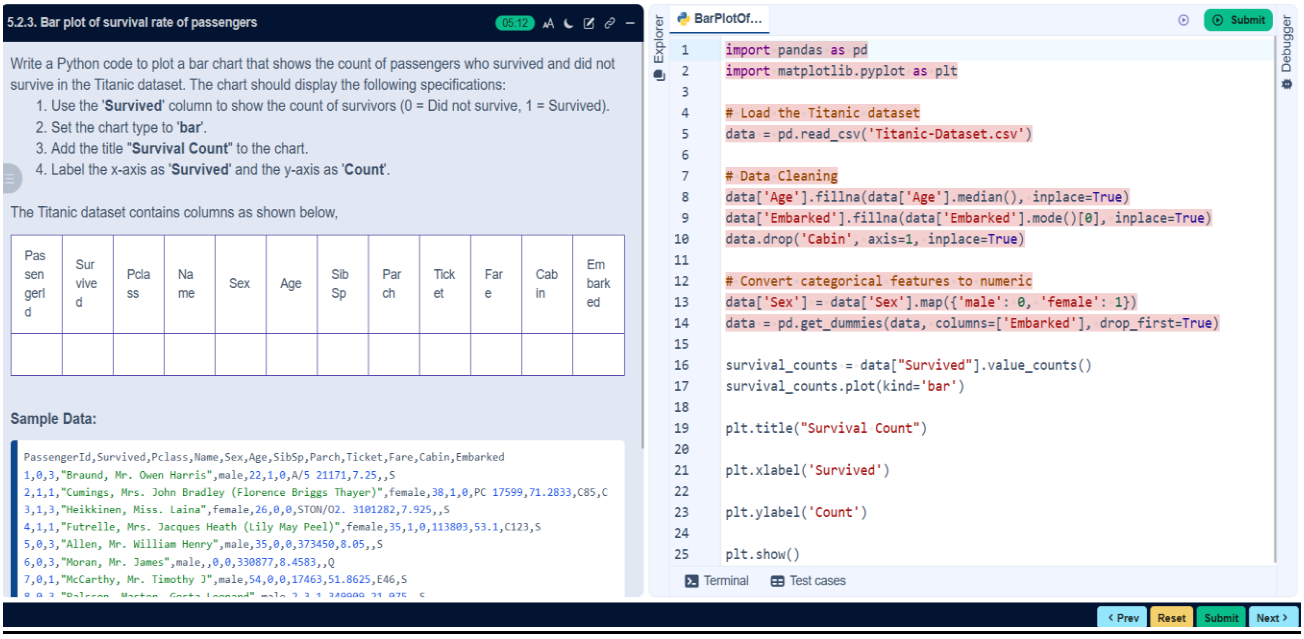


5.2.2 Histogram of passenger:

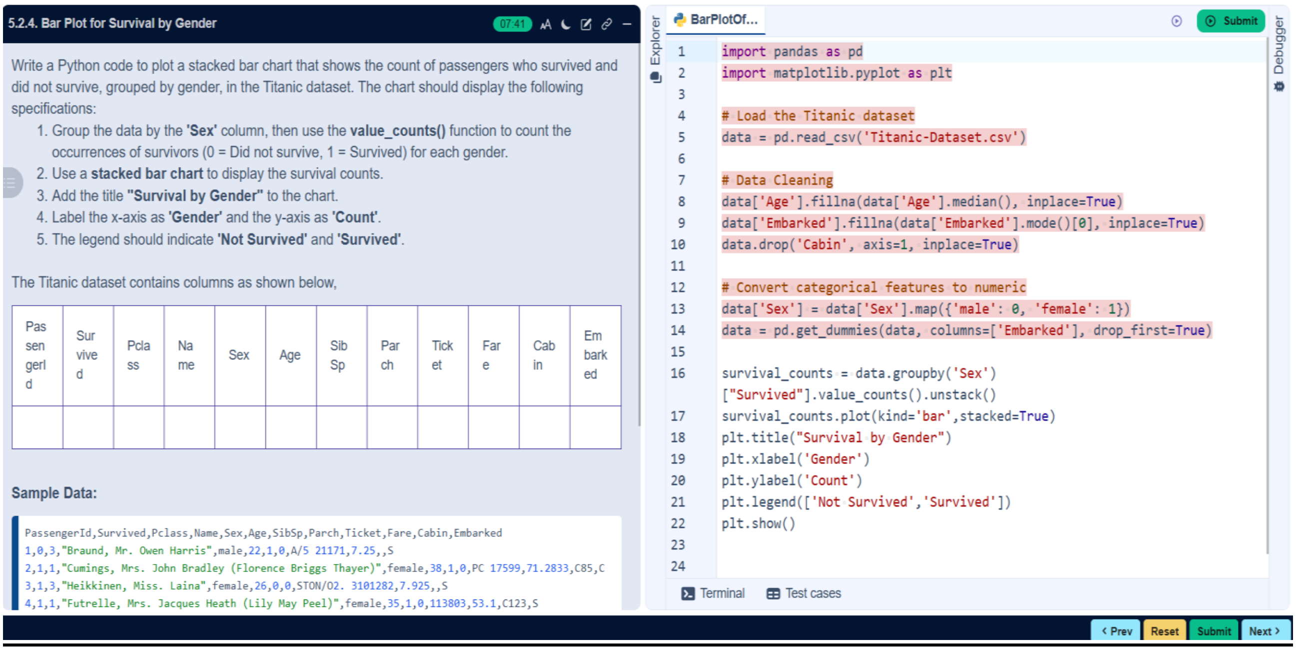




5.2.3 Bar plot of survival rate of passengers:



5.2.4 Bar plot for Survival of Gender:



5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **'Sex'** column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Gender"** to the chart.
4. Label the x-axis as **'Gender'** and the y-axis as **'Count'**.
5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

Passengerid	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

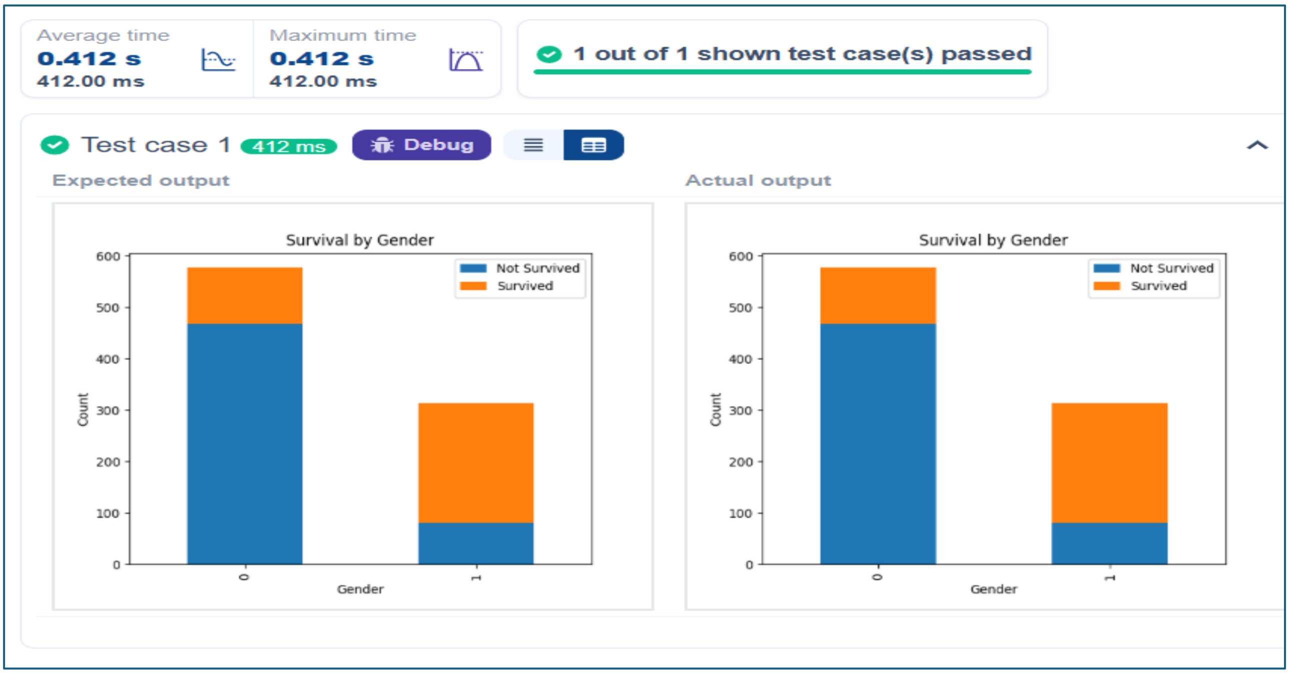
```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
```

BarPlotOf...

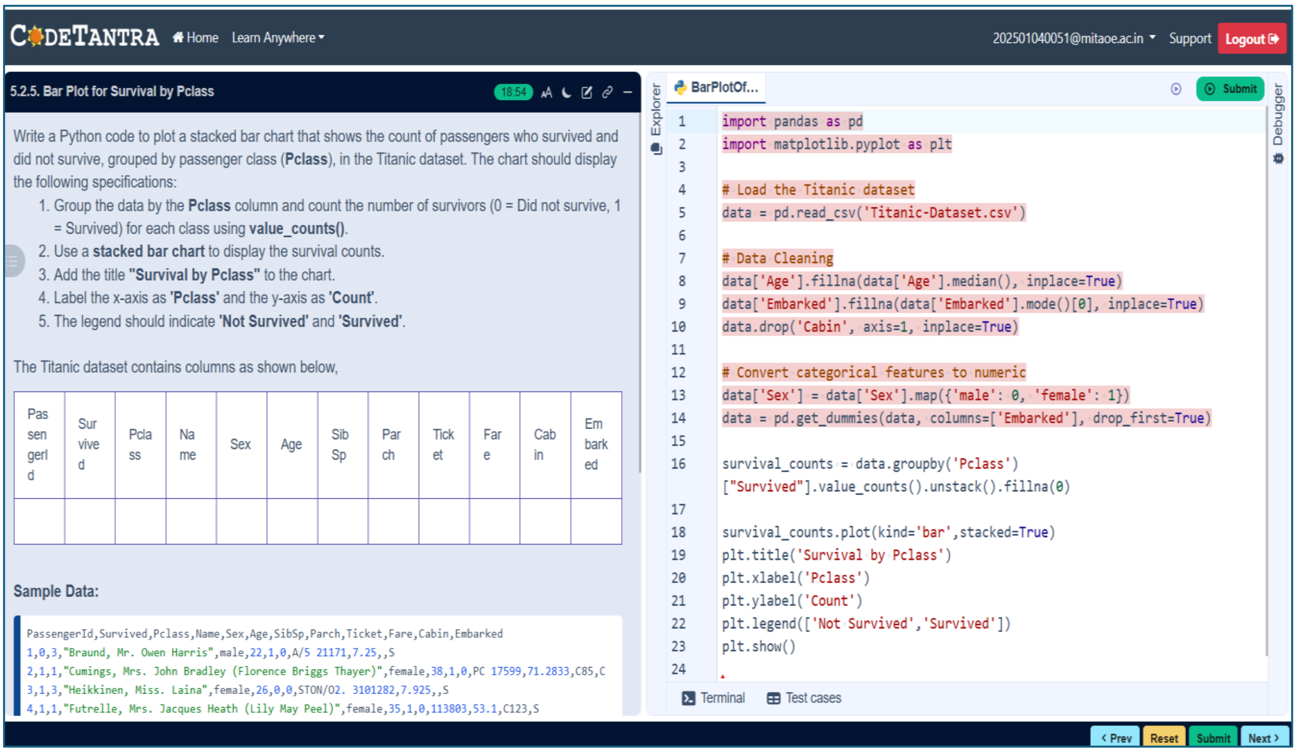
```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

survival_counts = data.groupby('Sex')
["Survived"].value_counts().unstack()
survival_counts.plot(kind='bar',stacked=True)
plt.title("Survival by Gender")
plt.xlabel('Gender')
plt.ylabel('Count')
plt.legend(['Not Survived','Survived'])
plt.show()
```

Terminal Test cases

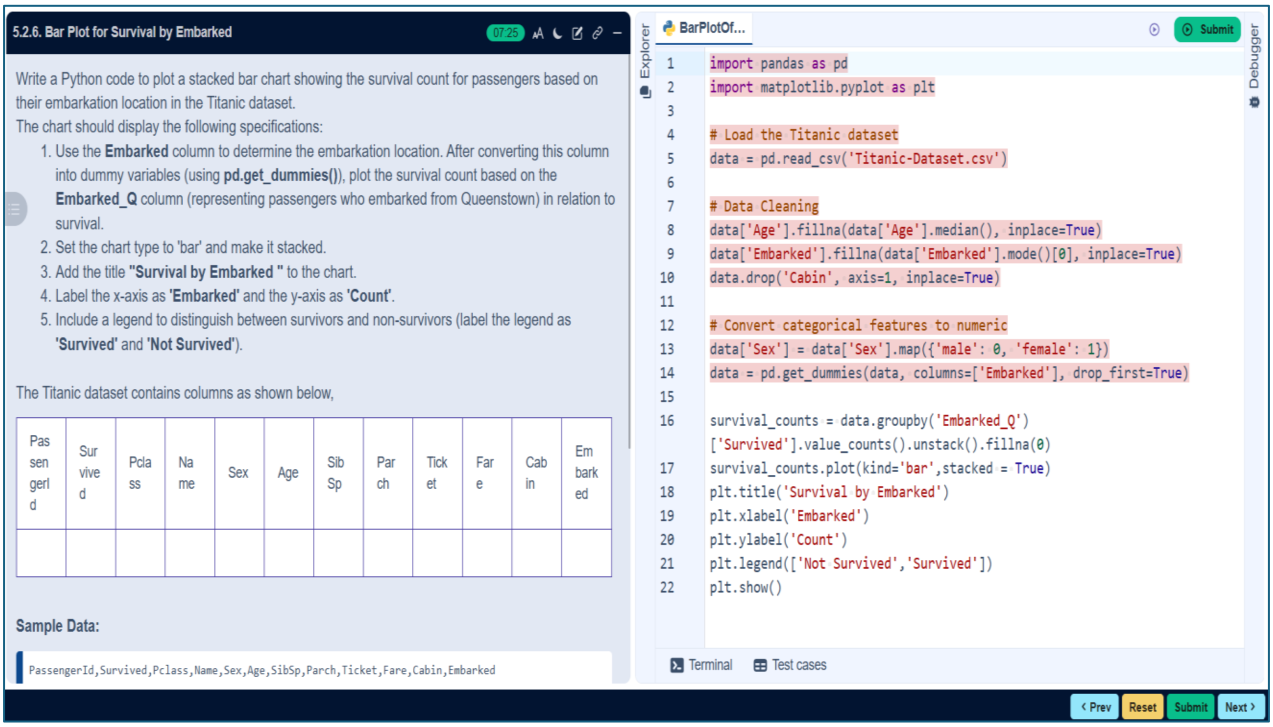
‹ Prev Reset Submit Next ›

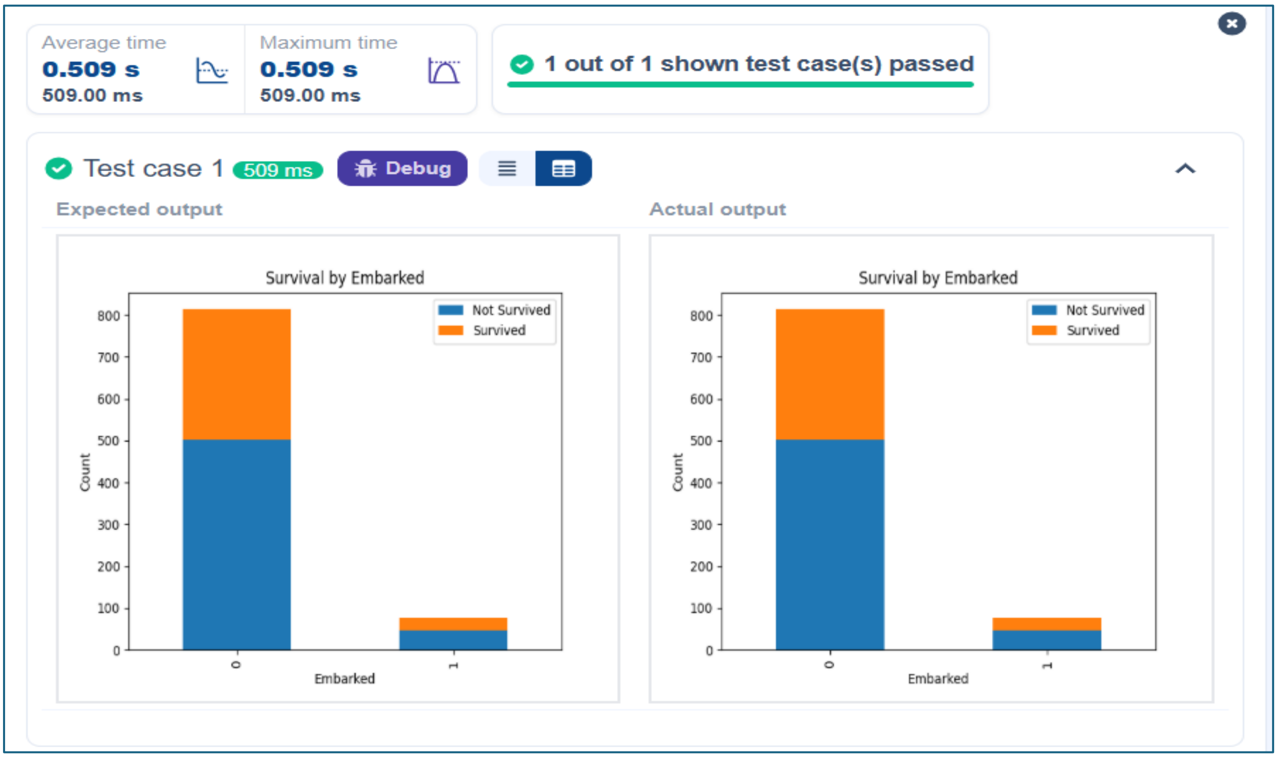


5.3.5 Bar plot survival of Pclass:

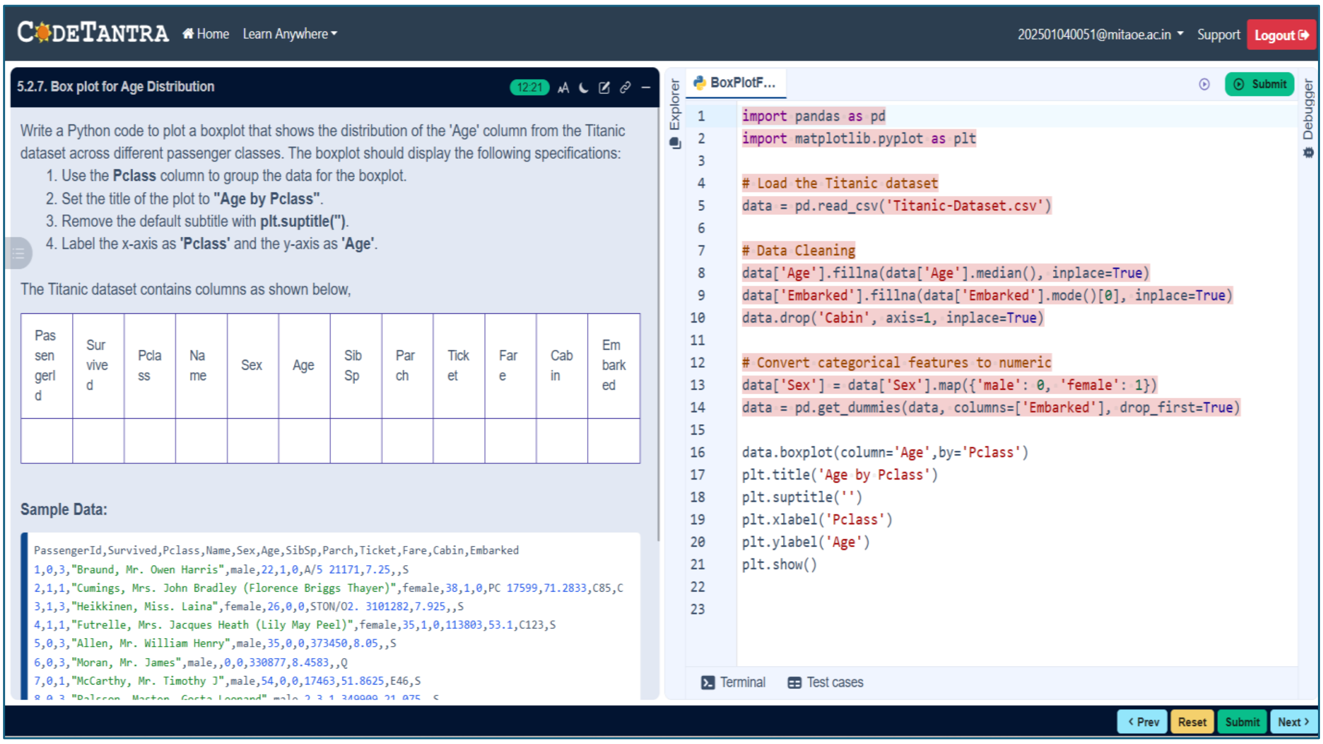


5.2.6 Bar Plot by Embarked:

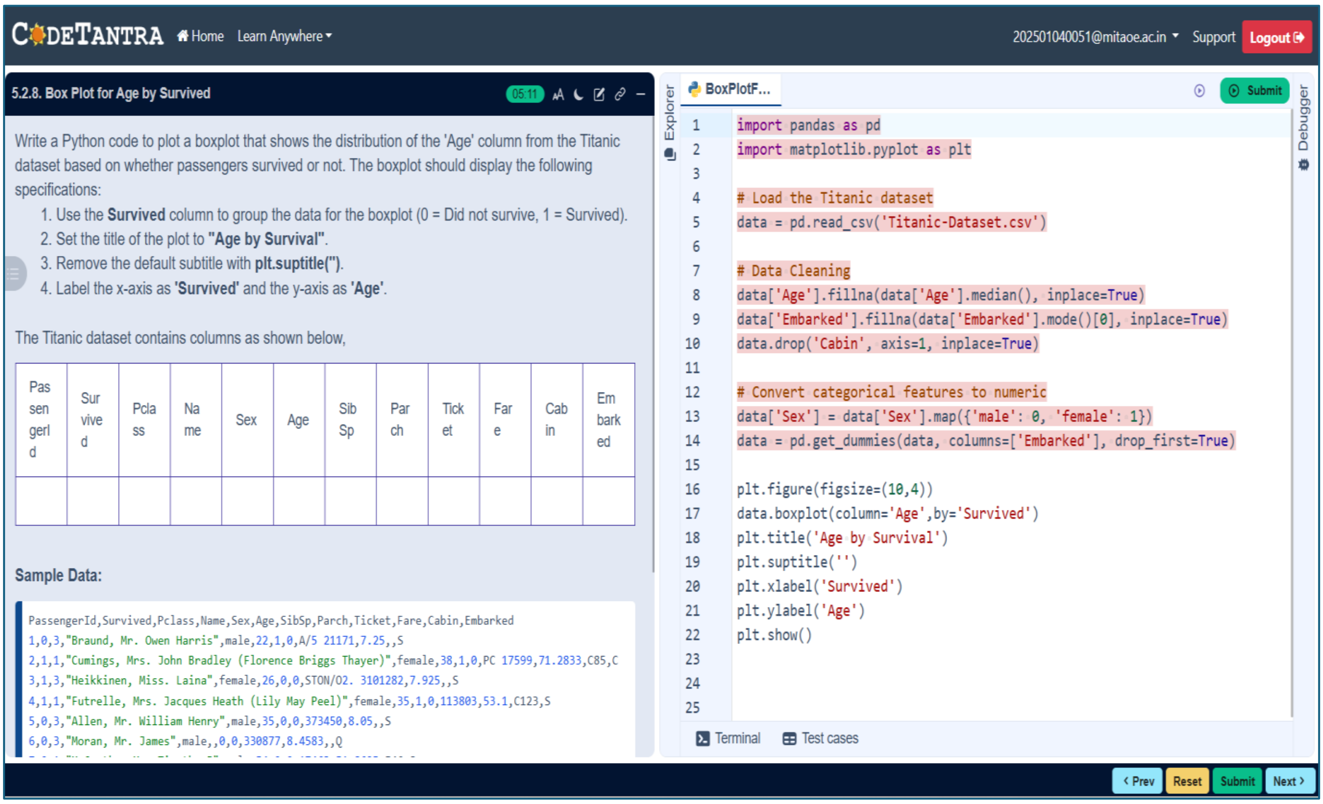


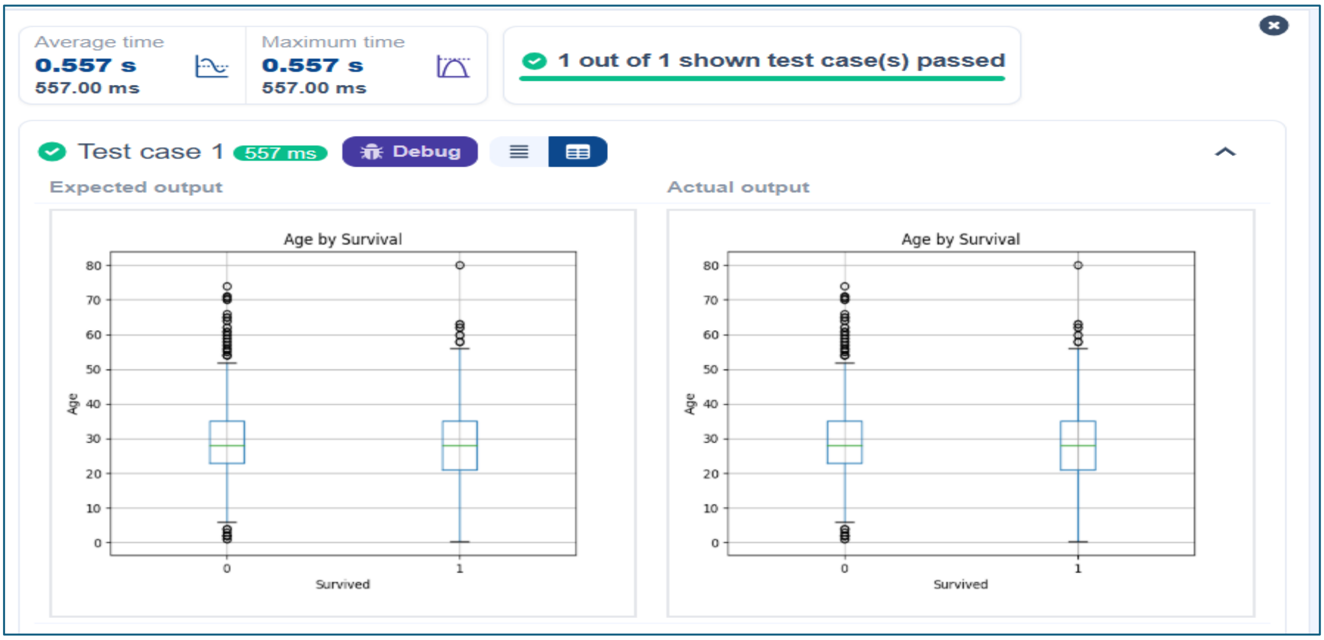


5.2.7 Box plot for Age Distribution:

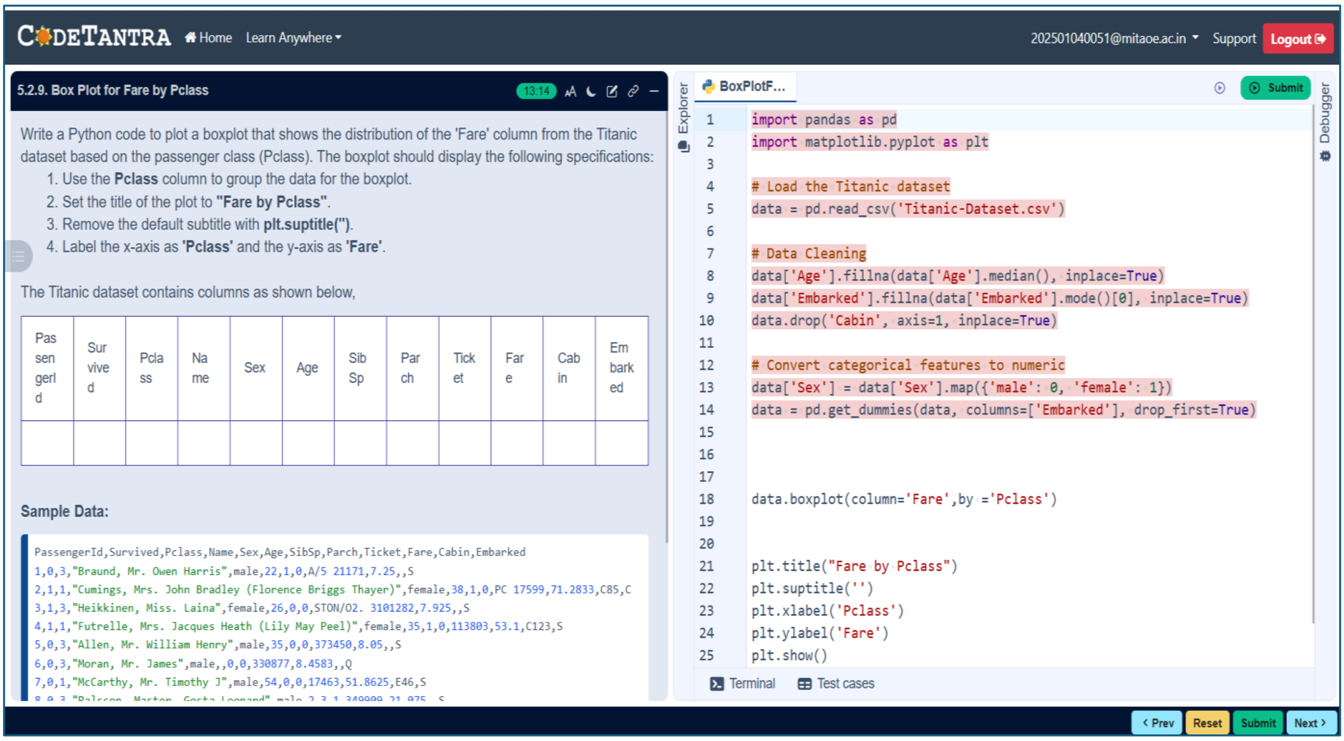


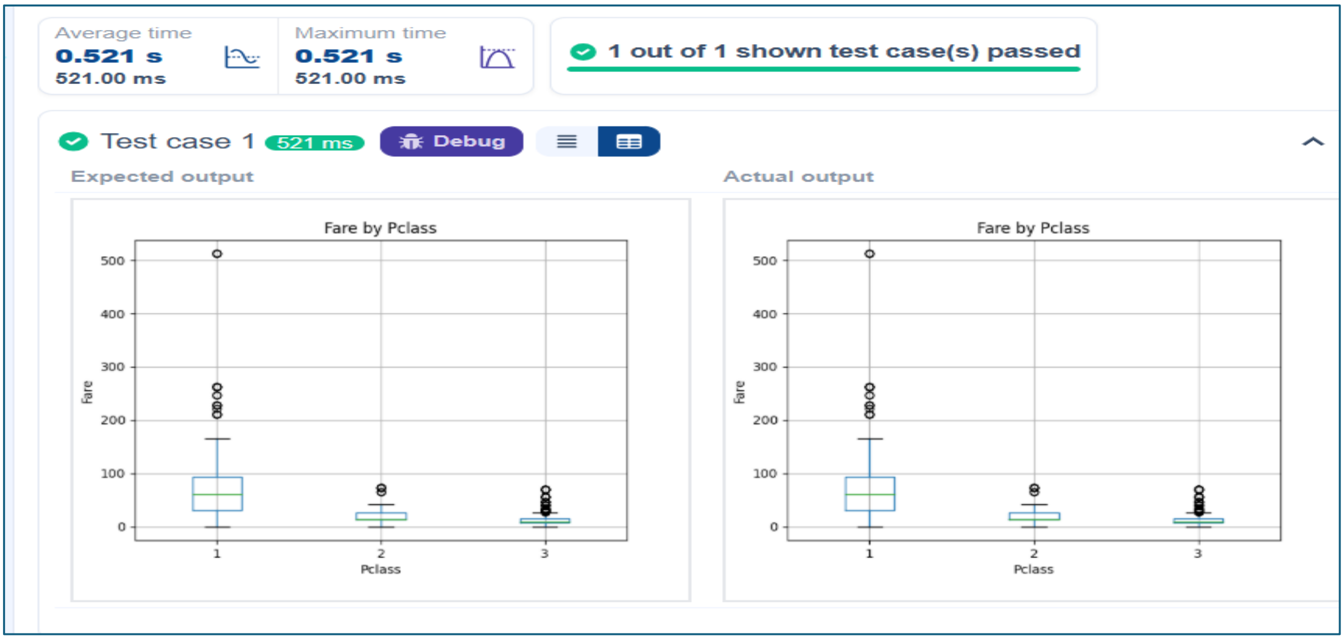
5.2.8 Box Plot for Age Survived:



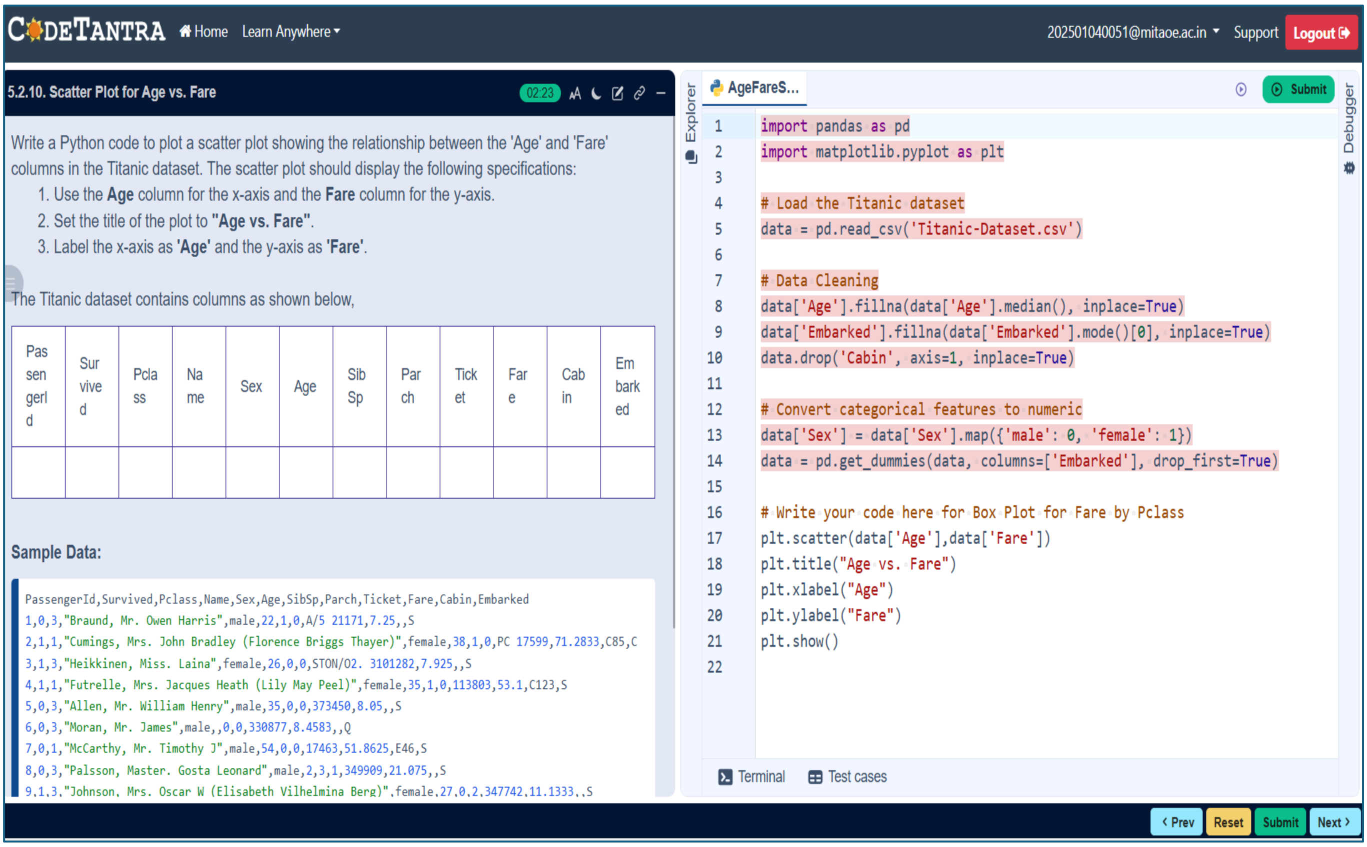


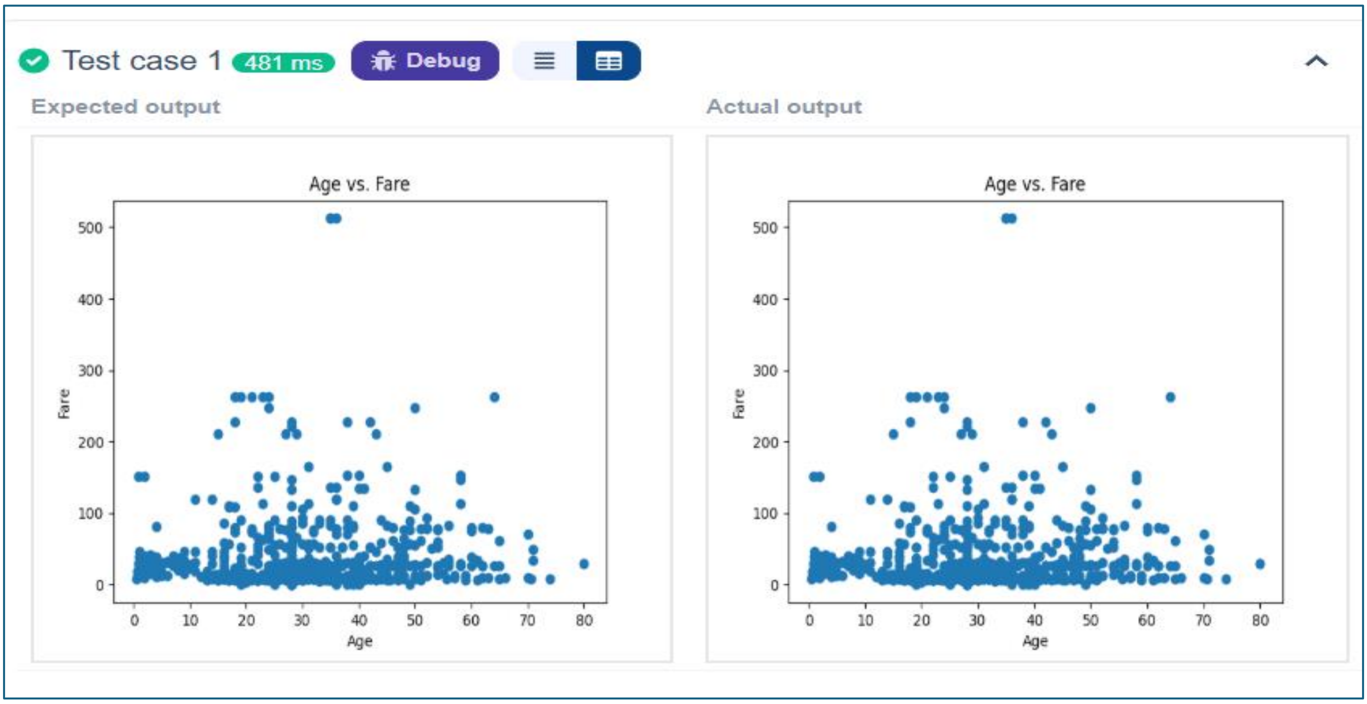
5.2.9 Box Plot for Fare by Pclass:





5.2.10 Scatter Plot for Age vs. Fare





5.2.11 Scatter Plot Age vs Fare by Survived: